Design and Implementation of an Autonomous 4WD Line-Following Robot

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Abstract— **This paper details the design and implementation of an autonomous four-wheel-drive line-following robot. Weighing approximately 1.5kg, the system is governed by a PIC18F47K42 microcontroller operating at 64MHz. The hardware architecture integrates an MX1508 motor driver, a GY-31 color sensor, an HC-SR04 ultrasonic sensor for obstacle detection, and an LM2596 buck converter for stable power distribution from a 7.4V 3000mAh battery. To ensure high-speed stability and reject sensor noise, the control software implements a non-linear Proportional-Integral-Derivative (PID) algorithm alongside custom three-stage confidence filters. These filters prevent false triggers by verifying line-loss and obstacle detection over multiple sequential readings. Furthermore, the robot utilizes a strict junction gatekeeper algorithm to navigate complex 24-node track layouts, enabling precise drift control, active braking, and targeted route merging.**

Keywords— Line Follower, PIC18F47K42, PID Control, Confidence Filter, Autonomous Navigation, Embedded Systems.

I. Introduction

Autonomous guided vehicles (AGVs) rely heavily on precise path tracking. As the mass and speed of the vehicle increase—such as in a 1.5kg four-wheel-drive (4WD) chassis—inertia becomes a significant factor that simple on-off control algorithms cannot handle. This project aims to solve the instability and false-reading issues inherent in high-speed line followers by utilizing a PIC18F47K42 microcontroller. By combining a tuned PID controller with multi-reading confidence filters for both ultrasonic and infrared sensor inputs, the robot achieves smooth track recovery and ignores corrupt data points.

II. Hardware Architecture

- Microcontroller Unit: The system is powered by the PIC18F47K42 configured with an internal high-frequency oscillator running at 64MHz, ensuring rapid polling of the sensor arrays and precise Pulse Width Modulation (PWM) generation. To ensure optimal polling speeds, the 5-point IR sensor array is sequentially mapped to the digital inputs of PORTB (RB0 to RB4). The HC-SR04 utilizes PORTC, where RC0 acts as the trigger and RC1 serves as the echo. The GY-31 colour sensor uses RA3, RA2, RE0, and RA5 as it's control pins; RA4 is the LED pin and RE1 is the output pin. The servo motor is also connected to RE2. A critical element of the design is the use of the microcontroller's Peripheral Pin Select (PPS) module. Rather than using standard digital pins to toggle motor direction, the software dynamically routes hardware PWM signals (PWM8 and PWM6) directly to PORTD. By locking and unlocking the PPS registers on the fly, the system achieves instant directional changes with embedded speed control.
- Drive System: A heavy 1.5kg 4WD chassis is driven by an MX1508 motor driver.
- Sensor Array: The robot utilizes a 5-point sensor array for line detection. Additionally, a GY-31 module is equipped for detecting red or blue and rotating the servo motor anti-clockwise or clockwise respectively, and an HC-SR04 ultrasonic sensor for obstacle detection on the nodes.
- Power Management: A 7-8V 3000mAh battery serves as the primary power source. To protect the logic-level sensors and the microcontroller from voltage spikes caused by motor stall currents, an LM2596 buck converter is utilized to step down and regulate the voltage.

III. Software Implementation

The logic is divided into three primary sub-systems to maintain stability on complex tracks:

- **Non-Linear PID Control:** The error is calculated based on a weighted non-linear scale ranging from -10 to +10 depending on the active sensors. The PID output, governed by the formula PID = (K_p . P) + (K_i . $\int e$) + (K_d . $\frac{de}{dt}$), adjusts the PWM duty cycle for the left and right motors. The tuning parameters are optimized to K_p = 30.0 and K_d = 200.0. It is important to note the fact that here the integral term (I) in a PID controller is often unnecessary, or even detrimental, for a Line Following Robot because the application rarely has a significant, persistent "steady-state error" that needs to be corrected. If the robot is forced to stop or if it takes a long time to navigate a turn, the I-term accumulates a massive error, causing the robot to over-correct and shoot off the line once it hits the next straight section.
- **Confidence Filtering Systems:** To mitigate false positives from sensor bounce or track reflections, the software utilizes a 3-reading confidence filter. For line recovery, if all sensors read zero, the system

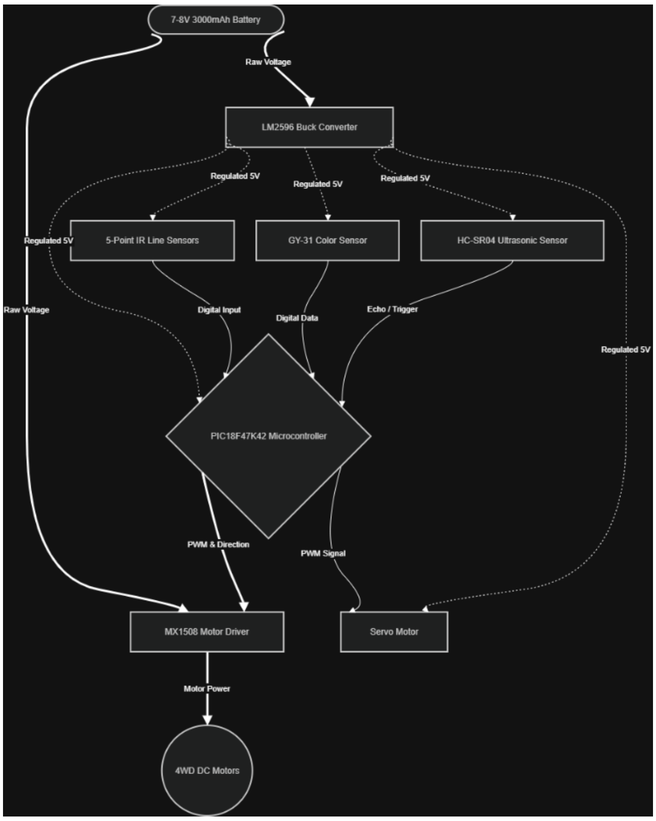
checks the array three times with a 1ms delay. Only if the confidence level is below the threshold does the system trigger a pivot recovery maneuver. Similarly, an ultrasonic confidence filter requires multiple consecutive readings below 6.5 cm to confirm a physical obstacle rather than a corrupt echo signal.

- **Junction Gatekeeper Protocol:** The track navigation is managed by a junction-counting latch mechanism. Depending on the current junction count (from 1 to 24), the robot executes distinct routines such as 180-degree U-turns, inertia-catch boosts, active braking, or left-wheel drifts.
- **Color-Based Decision and Actuation:** To enhance the robot's environmental awareness and enable interactive navigation, a GY-31 (TCS3200) color sensor is integrated alongside a servo motor mechanism. The software configures the sensor with a 20% frequency scaling to optimize the output pulse resolution for the microcontroller. By sequentially toggling the sensor's control pins (S2 and S3), the system isolates and reads the red, green, and blue (RGB) light intensities. Because the sensor outputs frequency proportional to light intensity, a shorter pulse duration indicates a stronger color presence.

  To filter out ambient light and prevent false triggers, the software employs a calibrated thresholding algorithm. The system first determines the dominant color by finding the minimum pulse width among the RGB values. It then verifies that this dominant value falls below a strict predefined threshold (THRESHOLD_VALUE = 900). Upon successful validation, the control logic triggers a specific physical response: detecting a valid red marker actuates the servo motor anticlockwise to 0 degrees, while a valid blue marker actuates it clockwise to 180 degrees. If the readings exceed the threshold or no clear color is dominant, the system recognizes it as ambient noise and returns the servo to a neutral 90-degree position.

IV. Diagrams

a. Block Diagram:



b. Flowchart: